



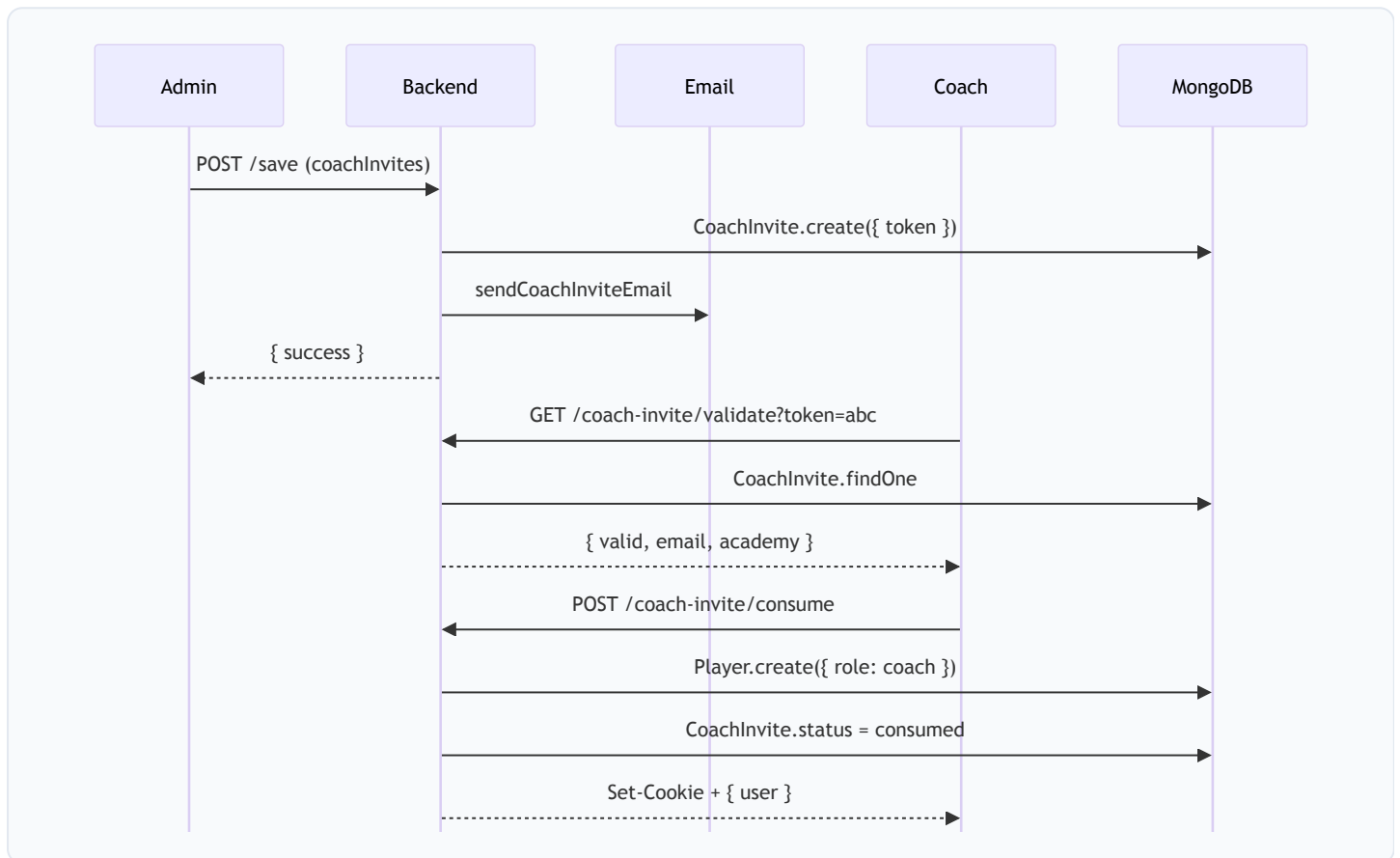
Coach Architecture

Complete system flow from the coach perspective — invite to evaluation

AceTrack Architecture Documentation • Generated 12 June 2026

1. Coach Onboarding Flow

Coaches are onboarded via an invite-only system managed by the Academy Admin. The invite link triggers a validation → account creation → login flow.



Detailed Flow Breakdown

1 Admin Creates Invite

The Admin saves a coach invite via the /save endpoint with the coach's email and sport expertise. The server generates a unique token, creates a CoachInvite document, and sends an onboarding email.

```
POST /api/v1/save { data: { coachInvites: [...] } }
```

2 Coach Validates Invite

Coach clicks the invite link. The app calls validate to check the token hasn't been consumed or expired. Returns the associated email and academy details.

```
GET /api/v1/auth/coach-invite/validate?token=abc123
```

3 Interaction Tracking

The system tracks invite analytics (link clicked, form opened) to give admins visibility into invite conversion rates.

```
POST /api/v1/auth/coach-invite/track { token, action }
```

4 Account Creation (Consume)

Coach submits their profile details and password. The server creates a Player document with role "coach", hashes the password, marks the invite as "consumed", and issues a JWT token.

```
POST /api/v1/auth/coach-invite/consume { token, name, password }
```

5 Subsequent Logins

After onboarding, coaches use the standard user login endpoint with username/password. The JWT cookie enables seamless session management.

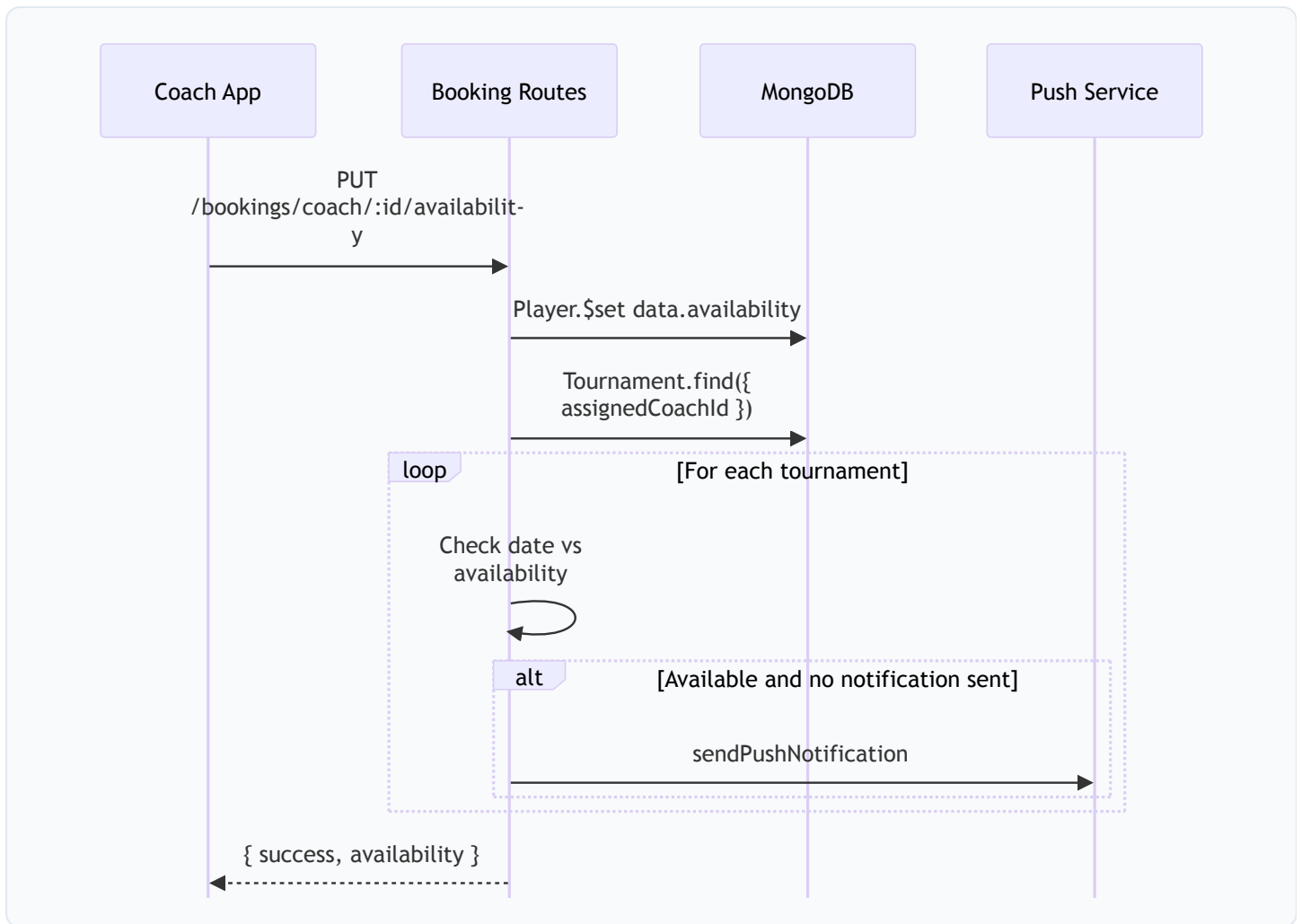
```
POST /api/v1/user/login { username, password }
```

API Endpoints

ENDPOINT	METHOD	PURPOSE
/auth/coach-invite/validate	GET	Validate invite token
/auth/coach-invite/track	POST	Track invite analytics
/auth/coach-invite/consume	POST	Create coach account
/admin/coach-invites	GET	List all invites (admin)
/user/login	POST	Standard login (post-onboarding)

2. Availability Management

Coaches set their weekly availability schedule, which determines tournament eligibility and triggers retroactive notifications for existing assignments.



Detailed Flow Breakdown

1 Set Weekly Schedule

Coach submits an array of availability slots, each with dayOfWeek (0-6), startTime, and endTime in 24-hour format. This replaces the entire availability array.

```
PUT /api/v1/bookings/coach/:coachId/availability { availability: [...] }
```

2 Retroactive Assignment Check

After updating availability, the server queries all tournaments where this coach is already assigned. For each tournament, it checks whether the event date/time falls within the new availability slots.

3 Availability Matching Logic

The system parses the tournament date to extract day-of-week and converts the event time to 24h format. It then checks if any availability slot matches both the day and time range.

4 Deferred Notifications

If a coach is assigned to a tournament that now falls within their availability, and they haven't been notified yet, the system sends a push notification: "Tournament Assignment 🏆 — You have been assigned as coach for [Tournament]".

5 Booking Status Updates

Coaches can accept, reject, or complete individual booking requests from players. Status transitions are: pending → confirmed → completed (or rejected).

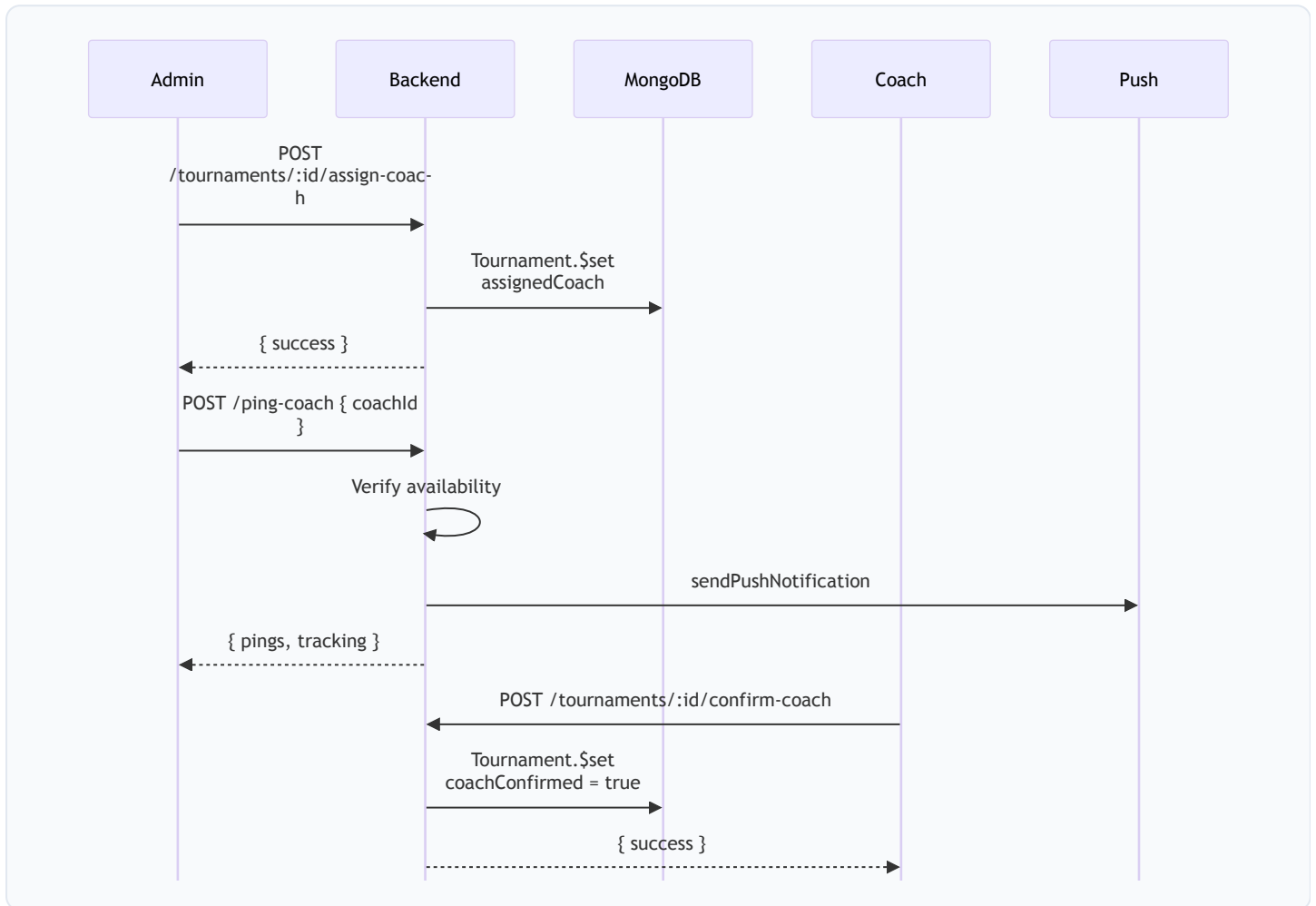
```
PUT /api/v1/bookings/:id/status { status }
```

API Endpoints

ENDPOINT	METHOD	PURPOSE
/bookings/coach/:coachId	GET	View incoming bookings
/bookings/:id/status	PUT	Accept/reject/complete booking
/bookings/coach/:coachId/availability	PUT	Set weekly availability

3. Tournament Assignment & Management

Admins assign coaches to tournaments, send push invites, and coaches confirm or decline assignments with optional comments.



Detailed Flow Breakdown

1 Admin Assigns Coach

Admin selects a coach for a tournament. The server updates the tournament's assignedCoach field and broadcasts an entity_updated event.

```
POST /api/v1/tournaments/:id/assign-coach { coachId }
```

2 Ping Coach (Push Invite)

Admin sends a push notification invite to the coach. The system first validates the coach is available at the tournament's date/time, then sends via Expo Push and tracks delivery status.

```
POST /api/v1/ping-coach { tournamentId, coachId }
```

3 Coach Confirms Assignment

Coach reviews the assignment details and confirms. This sets coachConfirmed: true on the tournament document.

```
POST /api/v1/tournaments/:id/confirm-coach
```

4 Coach Declines Assignment

If the coach declines, the assignedCoach field is cleared, allowing the admin to assign a different coach.

POST /api/v1/tournaments/:id/decline-coach

5 Coach Comment

Coach can add notes or instructions about the tournament (e.g., "Bring extra rackets", "Wet court conditions expected").

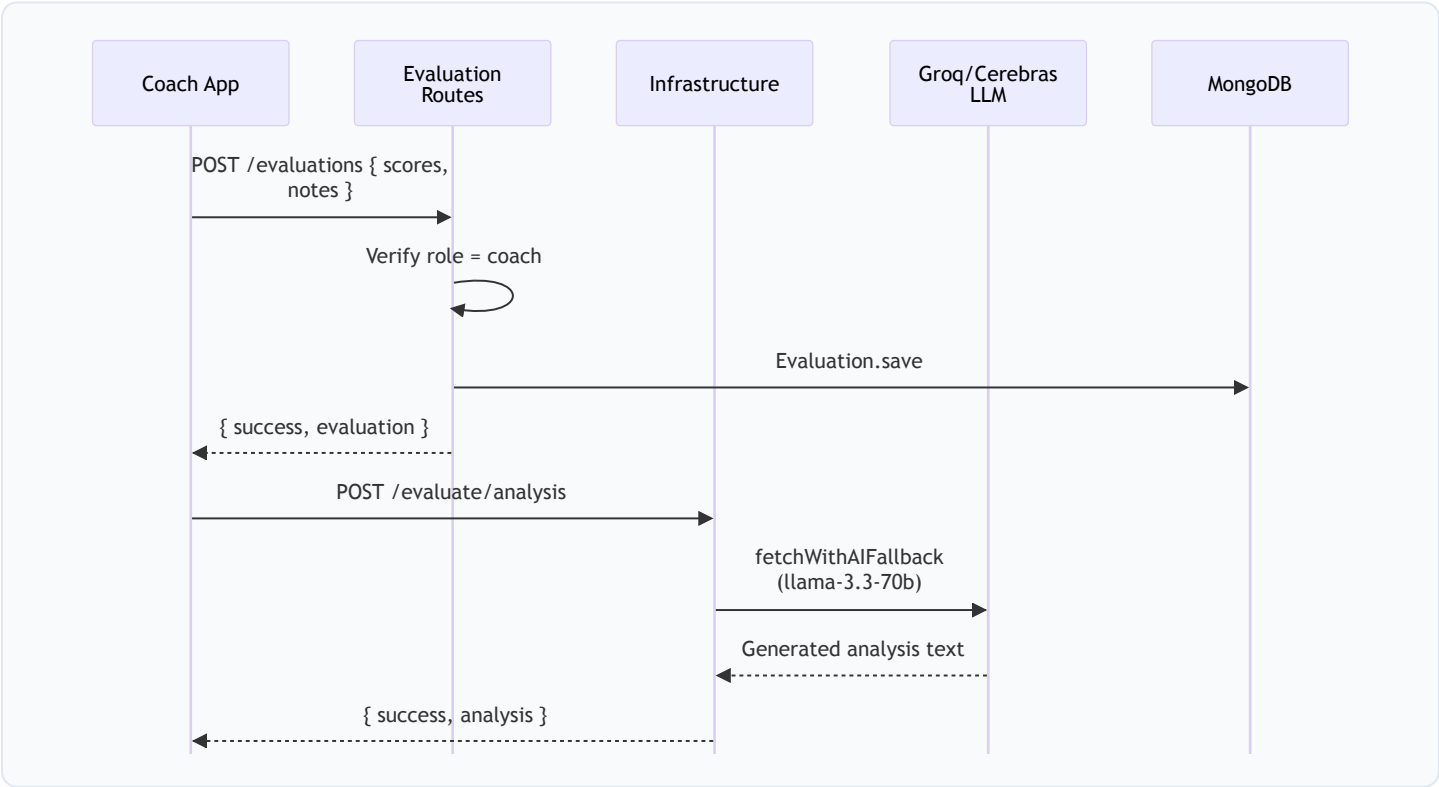
POST /api/v1/tournaments/:id/coach-comment { comment }

API Endpoints

ENDPOINT	METHOD	ROLE	PURPOSE
/tournaments/:id/confirm-coach	POST	Coach	Accept assignment
/tournaments/:id/decline-coach	POST	Coach	Decline assignment
/tournaments/:id/coach-comment	POST	Coach	Add notes
/tournaments/:id/assign-coach	POST	Admin	Assign coach
/ping-coach	POST	Admin	Send push invite

4. Player Evaluation & AI Analysis

Coaches evaluate player performance with structured scoring across multiple skill dimensions, with optional AI-powered analysis generation.



Detailed Flow Breakdown

1 Create Evaluation

Coach submits structured scores across skill dimensions: forehand, backhand, serve, footwork, strategy (each 0-100). Optional text notes provide qualitative feedback. The evaluatorId is enforced from JWT to prevent spoofing.

POST /api/v1/evaluations { playerId, scores, notes }

2 Ownership Enforcement

When updating an existing evaluation, the server verifies that doc.evaluatorId matches the requesting coach (or the user is an admin). This prevents coaches from editing each other's evaluations.

3 View Evaluations

Coach can list evaluations filtered by playerId, with results sorted by lastUpdated descending. Supports pagination with limit parameter.

```
GET /api/v1/evaluations?playerId=xyz&limit=50
```

4 AI Analysis Generation

Coach requests an AI-generated analysis based on the evaluation scores. The system uses Groq API with llama-3.3-70b-versatile model, with Cerebras as a fallback. The prompt instructs the AI to be encouraging and motivating.

```
POST /api/v1/infrastructure/evaluate/analysis { evaluationScores, playerSkillLevel }
```

5 Video Highlights

Coach can request AI-generated highlights from match video URLs. The infrastructure route processes the video and returns timestamps of key moments.

```
POST /api/v1/infrastructure/videos/:id/highlights { videoUrl }
```

API Endpoints

ENDPOINT	METHOD	PURPOSE
/evaluations	POST	Create/update evaluation
/evaluations	GET	List evaluations
/infrastructure/evaluate/analysis	POST	AI analysis from scores
/infrastructure/videos/:id/highlights	POST	Video highlight generation